LLM-Guided Drone Path Planning

Reference sheet for the 20-minute paper discussion — Arush Ram Anandh

Paper: Muhammad Sohail Farooq et al., *Goal-Oriented Intelligence Path Planning for LLMs Based Drone in Adversarial Conditions*. References below use page numbers in the supplied seven-page `paper.pdf`.

Research question and reported approach

Can foundation models use visual context and instructions to help a drone reach its goal while avoiding obstacles? The paper connects camera observations and YOLOv8 object detection to an LLM/VLM, then parses and interpolates generated waypoints before passing a trajectory through ROS to a motion controller. The reported evaluation uses **simulation**. [pp. 4–5, Fig. 3]

Terms used in the talk

LLM / VLM	Large language model / vision-language model; a VLM can process images alongside text.
Waypoint	An intermediate position along a proposed route.
YOLOv8	A learned object detector; detections alone do not supply 3D world coordinates.
RRT	Rapidly-exploring Random Tree: a classical randomized motion planner. Ordinary RRT does not guarantee a shortest path.
ROS	Robot Operating System: software infrastructure connecting robotics components.

Reported simulation results

Environment	Obstacles	Moving obstacles	Success rate	Mean time (s)
Simple	8–10	3	95.0%	65.7
Moderate	10–12	7	92.0%	65.4
Complex	12–16	10	91.5%	69.2
Dynamic	16–20	12	89.0%	72.0

[p. 5, Table I and Fig. 4] All four categories include moving obstacles. Trial counts, uncertainty, formal difficulty definitions and obstacle motion models are unspecified. Figure 4 calls these success percentages "efficiency" without defining a separate path-efficiency metric.

What the evidence leaves unresolved

- **Training and inference:** the paper describes both few-shot prompting and fine-tuning, but does not identify which system produced the aggregate results. Model identity, dataset size, splits and exact prompt are unclear.
- **Clearance and safety:** the text gives 2 m clearance on p. 4 and 1.2 m on p. 5. Interpolation adds points along segments; it alone ensures neither collision avoidance nor physically feasible flight.
- **Added value:** no controlled planner baseline, path-quality comparison or latency measurement establishes an LLM advantage. The displayed evaluation does not demonstrate physical flight, GPS-denied operation or bad-weather performance.

[pp. 4–6; evaluation gaps identified in this discussion]

Hybrid pipeline: our proposed extension

1. **LLM/VLM:** interpret the mission and propose high-level goals or constraints.
2. **Conventional planner:** compute a feasible route and check collisions.
3. **Controller:** track the trajectory.

This separates instruction understanding from geometric planning. **The paper does not evaluate this proposal**; its benefit and safety would need controlled testing.

Question to take away

What measurable capability does the foundation model add beyond the conventional planner used to generate its reference paths?